

AllianceMine Build & Release — Job Aid

Quick reference for running an AllianceMine build. For full details see ALLIANCEMINE_BUILD_AND_RELEASE.md.

Before You Start

```
# Check RDS free storage (need 250+ GB)
aws cloudwatch get-metric-statistics --namespace AWS/RDS \
  --metric-name FreeStorageSpace \
  --dimensions Name=DBInstanceIdentifier,Value=intermine-postgres \
  --start-time $(date -u -d '5 minutes ago' +%Y-%m-%dT%H:%M:%S) \
  --end-time $(date -u +%Y-%m-%dT%H:%M:%S) \
  --period 300 --statistics Average \
  --query 'Datapoints[0].Average' --output text | awk '{printf "%.0f GB\n",
  ↪ $1/1024/1024/1024}'

# Drop old RC databases if needed
PGPASSWORD='...' psql -h intermine-postgres.cmnhlso7wdi.us-east-1.rds.amazonaws.com \
  -U postgres -d postgres -c "SELECT datname, pg_size_pretty(pg_database_size(datname))
  FROM pg_database WHERE datname LIKE 'alliancemine_-' ORDER BY pg_database_size(datname)
  ↪ DESC"
```

Step 1: Configure

```
ssh -i ~/.ssh/AGR-ssl3.pem ec2-user@172.31.60.197
cd ~/agr_intermine_builder/docker/alliancemine
```

Edit .env — set ALLIANCE_RELEASE, RDS + SGD credentials. Solr URLs are auto-derived from SOLR_HOST + release + RC; the build will create alliancemine-search-{release}-rc{N} and the matching autocomplete core as a preflight step. Override SOLR_INDEX_URL / SOLR_AUTOCOMPLETE_URL only if you want to point at custom cores.

Step 2: Build Image + Compile

```
docker compose build                                # ~5 min
docker compose run -d --rm alliancemine-builder      # start for compilation
# Wait for compilation (~15 min):
CNAME=$(docker ps --format '{{.Names}}' | grep builder)
while docker exec $CNAME ls /root/.needs_compile 2>/dev/null; do sleep 30; done
docker commit $CNAME alliancemine-builder:9.0.0-compiled # save compiled image
docker stop $CNAME
```

Update docker-compose.yml -> image: alliancemine-builder:9.0.0-compiled

Step 3: Run Build

```
tmux new-session -s build
docker compose run --rm alliancemine-builder build # ~5-8 hours
```

Monitor: `tail -f /tmp/alliancemine-9.0.0-build.log`

Detach tmux: `Ctrl+B D` — Reattach: `tmux attach -t build`

Step 4: During Build — Manage Checkpoints

```
# List checkpoint databases
psql -h ... -d postgres -c \
  "SELECT datname, pg_size_pretty(pg_database_size(datname))
   FROM pg_database WHERE datname LIKE 'alliancemine_%rc:%'"

# Drop old ones (keep the latest)
psql -h ... -d postgres -c 'DROP DATABASE "alliancemine_9_0_0_rcN:old-checkpoint"'
```

Step 5: Fix Solr URLs + Rerun Indexes (if needed)

```
CNAME=$(docker ps --format '{{.Names}}' | grep builder)

# Patch hardcoded URLs
docker exec $CNAME sed -i \
  "s|index.solrurl = .*|index.solrurl =
  ↪ http://172.31.59.87:8983/solr/alliancemine-search-9.0.0|" \
  /root/alliancemine/dbmodel/resources/keyword_search.properties

docker exec $CNAME sed -i \
  "s|autocomplete.solrurl = .*|autocomplete.solrurl =
  ↪ http://172.31.59.87:8983/solr/alliancemine-autocomplete-9.0.0|" \
  /root/alliancemine/dbmodel/resources/objectstoresummary.config.properties

# Rerun indexes
docker exec $CNAME sh -c 'cd /root/alliancemine && ./gradlew postprocess
  ↪ -Pprocess=create-search-index --stacktrace'
docker exec $CNAME sh -c 'cd /root/alliancemine && ./gradlew postprocess
  ↪ -Pprocess=create-autocomplete-index --stacktrace'
```

Step 6: Deploy

```
# Start Tomcat container on multitenant
ssh -i ~/.ssh/AGR-ssl3.pem ec2-user@172.31.59.87
docker run -d --name alliancemine-9.0.0 \
  --add-host agr.stage.alliancemine.solr.server:host-gateway \
  -p 8082:8080 intermine-tomcat:latest
```

```
# Deploy WAR (from AllianceMineDev)
docker compose run --rm alliancemine-builder release \
  --tomcat-host 172.31.59.87 --tomcat-port 8082
```

Step 7: Verify

```
# Version endpoint
curl -s http://172.31.59.87:8082/alliancemine/service/version
# Expected: 35

# Solr cores
curl -s 'http://172.31.59.87:8983/solr/alliancemine-search-9.0.0/select?q=*&rows=0'
# Expected: numFound > 12,000,000
```

If Build Fails — Resume

```
docker compose run --rm -e RC_NUMBER=N alliancemine-builder build \
  --start-from project_build --resume
```

Uses `-l` flag to load last DB checkpoint. **Do not** combine with a fresh build (`-b` flag).

Key Numbers

Metric	Value
Full build time	5-8 hours
Main DB size	~57 GB
Each checkpoint	30-60 GB
RDS total storage	500 GB
Search index	~12.5M docs
Autocomplete index	~53K docs
JVM heap	48 GB
Container memory limit	56 GB
FMS data files	49
Data download	~16 GB

Emergency Contacts

System	Who to Contact
RDS storage full	Alliance DevOps
SGD database credentials	SGD team

System	Who to Contact
Solr down	Check multitenant host
NCBI Entrez failures	Retry — transient